

Assignment # 03: Loops (Iterative Structures)

Objective:

To help you understand and apply loops in Python Programming.

- Practice using **for** and **while** loops to repeat actions based on conditions..
 - Develop logical thinking.
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Instructions:

You are required to solve **each question using both types of loops**:

1. **First, solve the question using a `for` loop.**
2. **Then, solve the same question again using a `while` loop.**

For example:

Question: Print numbers from 1 to 30.

- Solution 1 → Using `for` loop
- Solution 2 → Using `while` loop

Make sure that both solutions produce the **same output**.

Loops (for and while)

1. Write a program to print numbers from 1 to 30 using for and while loops.
2. Write a program to print even numbers from 1 to 50 using for and while loops.
3. Write a program to print multiplication table of a given number for and while loops.
4. Write a program to print odd numbers from 1 to 50 using for and while loops.
5. Write a program to calculate the sum of all numbers from 1 to 50 using for and while loops
6. Write a program to print each character of a string for and while loops.
7. Write a program to compute the factorial of a number using for and while loops
8. Write a program to print numbers from 10 down to 1 using for and while loops.
[Hint: first check the sequence generated by range function on IDLE using this command i.e. **`list(range(10, 0, -1))`**]

9. Write a program that continues to ask user to input password until the correct password is entered using for and while loops.
10. Write a program to print the square of each number from 1 to 10 using for and while loops.
11. Write a program to calculate the sum of even and odd numbers and print them. (numbers from 1 to 50).
12. You are organizing a party and need to enter the names of **10 guests**. The program should check whether a guest's name has already been entered. Write a Python program that uses a for loop, list, input(), and if-else to manage a guest list and prevent duplicate names.
13. You are creating a restaurant menu. The program should allow the user to enter **5 dish names**. However, the restaurant does not serve **fish**, so the program should skip it whenever the user enters "fish". Write a Python program that uses a for loop, if condition, continue, and a list to create a restaurant menu while excluding a specific dish.
14. Aap ek **dawat ke liye menu aur uski cost calculate karne wala Python program** banayein. Dawat mein **300 mehman** hain. User se **5 dishes** ke naam lene hain. Lekin kuch dishes menu mein allowed nahi hain.

Requirements:

- discard_menu naam ki ek list banayein jismein ye dishes hon:
 1. fish
 2. qeema
 3. nihari
- loop ki madad se user se **5 dishes** ke naam input lein.
- Agar user aisi dish enter kare jo discard_menu mein mojud ho:
 1. Us dish ko menu mein add na karein.
 2. continue ka use karke next dish par chale jayein.
- Agar dish allowed ho to usay menu list mein add karein.
- Har allowed dish ke liye **300 mehmanon ka total price** user se input lein.
- Har dish ki price ko menu_price dictionary mein store karein.
- Loop complete hone ke baad tamam dishes ki prices ko add karke **total menu price** calculate karein.
- Har dish ki **per head cost** calculate karein:

$$\text{Per Head Cost} = \text{Total Dish Price} / \text{Number of Guests}$$

- Har dish ki per head cost ko menu_price_per_head dictionary mein store karein.
- Aakhir mein tamam dishes ki per head cost ko add karke **complete menu ki per head cost** display karein.